# BCG GenAI Consulting | Financial Statement Analysis

## Global Finance Corp. (GFC) — FY2023 to FY2025

**Forage Job Simulation | Task 1: Financial data extraction, cleaning and exploratory analysis**

**Coverage:** Microsoft, Tesla and Apple, three fiscal years per company (9 observations)
**Primary source:** U.S. Securities and Exchange Commission (SEC) EDGAR, Form 10-K
**Currency / units:** U.S. dollars, **millions (USD m)**, reported nominal figures
**Method:** Manual line-item extraction from public company filings, followed by reproducible Python/pandas analysis
**Deliverable:** A self-contained, executed Jupyter Notebook with financial trends, data quality checks, visualizations, and recommendations for an AI financial chatbot.

> **Simulation context:** This is an educational BCG Forage job-simulation deliverable, not an official BCG research publication or investment recommendation.

### 1 | Executive overview

We examine five financial line items across FY2023–FY2025:

1. **Total revenue:** Annual sales / total net sales.
2. **Net income:** Consolidated net income reported on the income statement.
3. **Total assets:** Balance-sheet assets at fiscal year end.
4. **Total liabilities:** Balance-sheet obligations at fiscal year end.
5. **Operating cash flow (OCF):** Net cash generated by operating activities.

The analysis calculates year-over-year changes for **each of the five metrics**, compares profit and cash-generation margins, and highlights important differences in financial trajectory. All findings are historical; the project does **not** forecast stock prices.

## 2 | Source research and extraction protocol

**SEC EDGAR 10-K filings reviewed:**

| Company | FY2023 Form 10-K | FY2024 Form 10-K | FY2025 Form 10-K |
|---|---|---|---|
| **Microsoft** | 2023 10-K | 2024 10-K | 2025 10-K |
| **Tesla** | 2023 10-K | 2024 10-K | 2025 10-K |
| **Apple** | 2023 10-K | 2024 10-K | 2025 10-K |

**Line-item extraction:** "Total revenue" / "Total revenues" / "Total net sales" and "Net income" were taken from **Item 8 income statements**; "Total assets" and "Total liabilities" from **Item 8 balance sheets**; "Net cash from operations" / "Net cash provided by operating activities" / "Cash generated by operating activities" from **Item 8 cash flow statements**.

**Audit trail:** Each source URL is retained in the raw data. The FY2025 10-K contains FY2023, FY2024 and FY2025 income/cash-flow columns, but generally only FY2024 and FY2025 balance sheets. Thus **FY2023 balance-sheet values are sourced from the FY2024 10-K**. A separate filing link for each company's fiscal year is listed above.

**Definition note:** Tesla's **net income** is the consolidated line (FY2023 14,974; FY2024 7,153; FY2025 3,855 USD m), **not** the different "net income attributable to common stockholders" line. This keeps the requested metric consistent.

**Accounting/time note:** Microsoft's fiscal year ends June 30; Apple's ends in late September; Tesla's ends December 31. FY2025 figures therefore cover **different twelve-month windows**. This is a directional benchmark, not a perfectly aligned calendar-year comparison. Tesla's FY2025 10-K was filed in 2026, after the fiscal year ended.

## 3 | Set up Python and import the source data

To keep this submission **upload-ready as a single `.ipynb` file**, the verified source CSV is embedded below. It is read using `pandas.read_csv`, so the notebook works even if the separate CSV or Excel attachment is unavailable. The same data is also provided as accompanying downloadable files.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from io import StringIO
from IPython.display import display

pd.set_option('display.max_columns', 20)
pd.set_option('display.width', 180)

print('Libraries imported successfully.')
```

```
Libraries imported successfully.
```

In [2]:

```
# Embedded manual extraction from official SEC 10-K financial statements.
# Values are numeric USD millions, WITHOUT $ signs or thousands separators.
# Source URLs are retained for traceability.
raw_csv = r'''Company,Fiscal Year,Fiscal Year End,Total Revenue (USD millions),Net Income
(USD millions),Total Assets (USD millions),Total Liabilities (USD millions),Operating Cash
Flow (USD millions),Financial Statement Source (10-K),Balance Sheet Source (10-K)
Microsoft,2023,2023-06-
30,211915,72361,411976,205753,87582,https://www.sec.gov/Archives/edgar/data/789019/000095017
025100235/msft-
20250630.htm,https://www.sec.gov/Archives/edgar/data/789019/000095017024087843/msft-
20240630.htm
Microsoft,2024,2024-06-
30,245122,88136,512163,243686,118548,https://www.sec.gov/Archives/edgar/data/789019/00009501
7025100235/msft-
20250630.htm,https://www.sec.gov/Archives/edgar/data/789019/000095017025100235/msft-
20250630.htm
Microsoft,2025,2025-06-
30,281724,101832,619003,275524,136162,https://www.sec.gov/Archives/edgar/data/789019/0000950
17025100235/msft-
20250630.htm,https://www.sec.gov/Archives/edgar/data/789019/000095017025100235/msft-
20250630.htm
Tesla,2023,2023-12-
31,96773,14974,106618,43009,13256,https://www.sec.gov/Archives/edgar/data/1318605/0001628280
26003952/tsla-
20251231.htm,https://www.sec.gov/Archives/edgar/data/1318605/000162828025003063/tsla-
20241231.htm
Tesla,2024,2024-12-
31,97690,7153,122070,48390,14923,https://www.sec.gov/Archives/edgar/data/1318605/00016282802
6003952/tsla-
20251231.htm,https://www.sec.gov/Archives/edgar/data/1318605/000162828026003952/tsla-
20251231.htm
Tesla,2025,2025-12-
31,94827,3855,137806,54941,14747,https://www.sec.gov/Archives/edgar/data/1318605/00016282802
6003952/tsla-
20251231.htm,https://www.sec.gov/Archives/edgar/data/1318605/000162828026003952/tsla-
20251231.htm
Apple,2023,2023-09-
30,383285,96995,352583,290437,110543,https://www.sec.gov/Archives/edgar/data/320193/00003201
9325000079/aapl-
20250927.htm,https://www.sec.gov/Archives/edgar/data/320193/000032019324000123/aapl-
20240928.htm
Apple,2024,2024-09-
28,391035,93736,364980,308030,118254,https://www.sec.gov/Archives/edgar/data/320193/00003201
9325000079/aapl-
20250927.htm,https://www.sec.gov/Archives/edgar/data/320193/000032019325000079/aapl-
20250927.htm
Apple,2025,2025-09-
27,416161,112010,359241,285508,111482,https://www.sec.gov/Archives/edgar/data/320193/0000320
19325000079/aapl-
20250927.htm,https://www.sec.gov/Archives/edgar/data/320193/000032019325000079/aapl-
20250927.htm
'''
df = pd.read_csv(StringIO(raw_csv.strip()))

df = df.sort_values(['Company', 'Fiscal Year']).reset_index(drop=True)

print(f'Data loaded: {df.shape[0]} rows and {df.shape[1]} columns')
display(df.iloc[:, :8].style.format({
    'Total Revenue (USD millions)': '{:,.0f}',
    'Net Income (USD millions)': '{:,.0f}',
    'Total Assets (USD millions)': '{:,.0f}',
    'Total Liabilities (USD millions)': '{:,.0f}',
    'Operating Cash Flow (USD millions)': '{:,.0f}',
}))
```

```
Data loaded: 9 rows and 10 columns
```

| | Company | Fiscal Year | Fiscal Year End | Total Revenue (USD millions) | Net Income (USD millions) | Total Assets (USD millions) | Total Liabilities (USD millions) | Operating Cash Flow (USD millions) |
|---|---|---|---|---|---|---|---|---|
| **0** | Apple | 2023 | 2023-09-30 | 383,285 | 96,995 | 352,583 | 290,437 | 110,543 |
| **1** | Apple | 2024 | 2024-09-28 | 391,035 | 93,736 | 364,980 | 308,030 | 118,254 |
| **2** | Apple | 2025 | 2025-09-27 | 416,161 | 112,010 | 359,241 | 285,508 | 111,482 |
| **3** | Microsoft | 2023 | 2023-06-30 | 211,915 | 72,361 | 411,976 | 205,753 | 87,582 |
| **4** | Microsoft | 2024 | 2024-06-30 | 245,122 | 88,136 | 512,163 | 243,686 | 118,548 |
| **5** | Microsoft | 2025 | 2025-06-30 | 281,724 | 101,832 | 619,003 | 275,524 | 136,162 |
| **6** | Tesla | 2023 | 2023-12-31 | 96,773 | 14,974 | 106,618 | 43,009 | 13,256 |
| **7** | Tesla | 2024 | 2024-12-31 | 97,690 | 7,153 | 122,070 | 48,390 | 14,923 |
| **8** | Tesla | 2025 | 2025-12-31 | 94,827 | 3,855 | 137,806 | 54,941 | 14,747 |

## 4 | Cleaning and validation

Raw values are already standardized to **USD millions**, with no currency symbols, commas, negative numbers disguised as text or missing cells. We nevertheless run explicit data-quality controls for reproducibility:

- confirm exactly three companies × three years (9 unique company-year combinations);
- check missing/duplicate records and numeric types;
- verify nonnegative values and basic balance-sheet reasonableness (liabilities do not exceed assets in this sample);
- preserve fiscal year-end dates and original SEC source URLs, rather than guessing dates.

> Different fiscal-year-end dates are **not** converted to calendar years, and reported nominal USD amounts are **not** inflation-adjusted. Such a conversion would be an assumption, not manual extraction.

In [3]:
```
numeric_metrics = [
    'Total Revenue (USD millions)',
    'Net Income (USD millions)',
    'Total Assets (USD millions)',
    'Total Liabilities (USD millions)',
    'Operating Cash Flow (USD millions)',
]

# Enforce consistent data types.
df['Fiscal Year'] = pd.to_numeric(df['Fiscal Year'], errors='raise').astype(int)
df['Fiscal Year End'] = pd.to_datetime(df['Fiscal Year End'], errors='raise')
for field in numeric_metrics:
    df[field] = pd.to_numeric(df[field], errors='raise')

checks = {
    'All 9 company-year observations present': len(df) == 9 and df[['Company','Fiscal
Year']].drop_duplicates().shape[0] == 9,
    'Three fiscal years per company': (df.groupby('Company')['Fiscal Year'].nunique() ==
3).all(),
    'No missing source fields or numeric observations': df.isna().sum().sum() == 0,
    'No negative extracted metrics': (df[numeric_metrics] >= 0).all().all(),
    'Reported liabilities do not exceed assets': (df['Total Liabilities (USD millions)'] <=
df['Total Assets (USD millions)']).all(),
    'Fiscal-year labels match year-end dates': (df['Fiscal Year End'].dt.year == df['Fiscal
Year']).all(),
}
quality_report = pd.DataFrame([{'Check': k, 'Result': 'PASS' if v else 'FAIL'} for k,v in
checks.items()])
display(quality_report)
assert all(checks.values()), 'Data validation failed: inspect the extraction before
proceeding.'
print('All data-quality checks passed.')
```

| | Check | Result |
|---|---|---|
| **0** | All 9 company-year observations present | PASS |
| **1** | Three fiscal years per company | PASS |
| **2** | No missing source fields or numeric observations | PASS |
| **3** | No negative extracted metrics | PASS |
| **4** | Reported liabilities do not exceed assets | PASS |
| **5** | Fiscal-year labels match year-end dates | PASS |

```
All data-quality checks passed.
```

## 5 | Calculate annual change for ALL five metrics

For each company and each financial measure:

[\text{Year-over-Year Change (%)} = \frac{\text{Current fiscal year} - \text{Previous fiscal year}}{\text{Previous fiscal year}}\times100]

The first year (2023) is intentionally **N/A** because no FY2022 baseline was extracted for this exercise. Company grouping and sorting ensure each percentage compares the **same company** with its previous fiscal year.

In [4]:
```
df = df.sort_values(['Company', 'Fiscal Year']).reset_index(drop=True)

short_names = {
    'Total Revenue (USD millions)': 'Revenue',
    'Net Income (USD millions)': 'Net Income',
    'Total Assets (USD millions)': 'Total Assets',
    'Total Liabilities (USD millions)': 'Total Liabilities',
    'Operating Cash Flow (USD millions)': 'Operating Cash Flow',
}

for original, shorter in short_names.items():
    df[f'{shorter} YoY (%)'] = df.groupby('Company')[original].pct_change(fill_method=None)
* 100

change_cols = ['Company','Fiscal Year'] + [f'{k} YoY (%)' for k in short_names.values()]
display(df[change_cols].style.format({c: '{:+.2f}%' for c in change_cols[2:]},
na_rep='N/A'))
```

| | Company | Fiscal Year | Revenue YoY (%) | Net Income YoY (%) | Total Assets YoY (%) | Total Liabilities YoY (%) | Operating Cash Flow YoY (%) |
|---|---|---|---|---|---|---|---|
| **0** | Apple | 2023 | N/A | N/A | N/A | N/A | N/A |
| **1** | Apple | 2024 | +2.02% | -3.36% | +3.52% | +6.06% | +6.98% |
| **2** | Apple | 2025 | +6.43% | +19.50% | -1.57% | -7.31% | -5.73% |
| **3** | Microsoft | 2023 | N/A | N/A | N/A | N/A | N/A |
| **4** | Microsoft | 2024 | +15.67% | +21.80% | +24.32% | +18.44% | +35.36% |
| **5** | Microsoft | 2025 | +14.93% | +15.54% | +20.86% | +13.07% | +14.86% |
| **6** | Tesla | 2023 | N/A | N/A | N/A | N/A | N/A |
| **7** | Tesla | 2024 | +0.95% | -52.23% | +14.49% | +12.51% | +12.58% |
| **8** | Tesla | 2025 | -2.93% | -46.11% | +12.89% | +13.54% | -1.18% |

## 6 | Financial health indicators (feature engineering)

We calculate simple, interpretable chatbot-ready indicators:

- **Net profit margin** = net income ÷ total revenue × 100.
- **Operating cash flow margin** = operating cash flow ÷ revenue × 100.
- **Liabilities-to-assets** = total liabilities ÷ total assets × 100. *(A book-balance-sheet funding measure, not an interest-bearing debt ratio.)*
- **Operating cash / net income** = operating cash flow ÷ net income. *(A cash-conversion indicator, with important working-capital and noncash-item caveats.)*

These describe reported financial outcomes, not investment valuations or forward-looking risk forecasts.

In [5]:
```
df['Net Profit Margin (%)'] = 100 * df['Net Income (USD millions)'] / df['Total Revenue (USD millions)']
df['Operating Cash Flow Margin (%)'] = 100 * df['Operating Cash Flow (USD millions)'] / df['Total Revenue (USD millions)']
df['Liabilities-to-Assets (%)'] = 100 * df['Total Liabilities (USD millions)'] / df['Total Assets (USD millions)']
df['Operating Cash / Net Income (x)'] = df['Operating Cash Flow (USD millions)'] / df['Net Income (USD millions)']

snapshot = df.loc[df['Fiscal Year']==2025, [
    'Company','Total Revenue (USD millions)','Net Income (USD millions)',
    'Operating Cash Flow (USD millions)','Net Profit Margin (%)',
    'Operating Cash Flow Margin (%)', 'Liabilities-to-Assets (%)',
    'Operating Cash / Net Income (x)'
]].sort_values('Company')

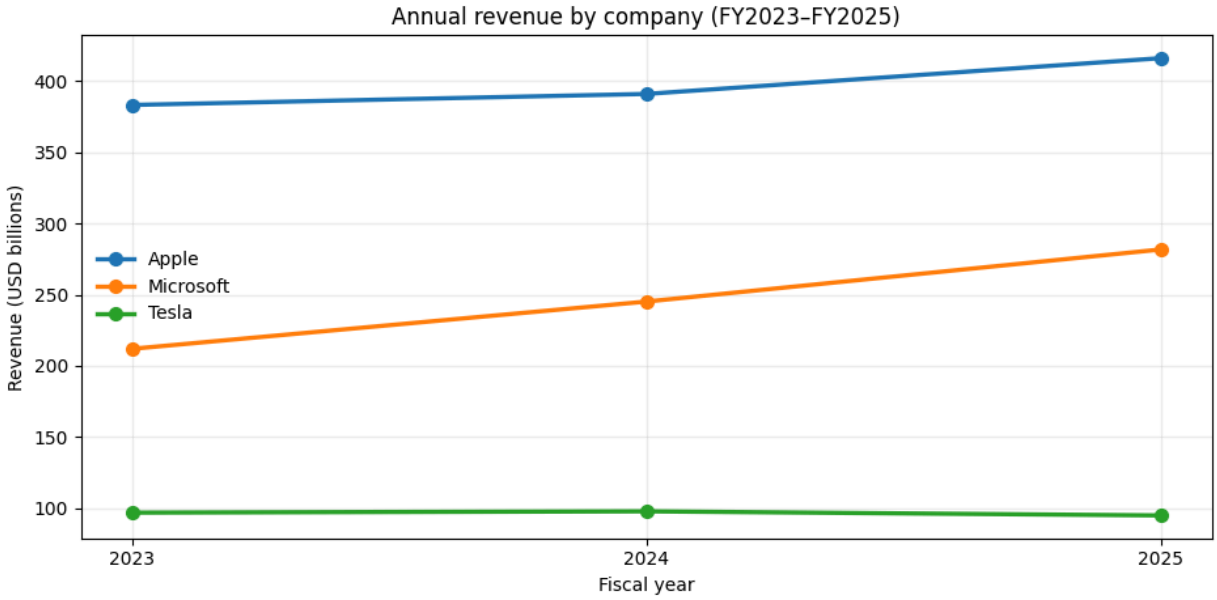
display(snapshot.style.format({
    'Total Revenue (USD millions)': '{:,.0f}',
    'Net Income (USD millions)': '{:,.0f}',
    'Operating Cash Flow (USD millions)': '{:,.0f}',
    'Net Profit Margin (%)': '{:.2f}%',
    'Operating Cash Flow Margin (%)': '{:.2f}%',
    'Liabilities-to-Assets (%)': '{:.2f}%',
    'Operating Cash / Net Income (x)': '{:.2f}x',
}))
```

| | Company | Total Revenue (USD millions) | Net Income (USD millions) | Operating Cash Flow (USD millions) | Net Profit Margin (%) | Operating Cash Flow Margin (%) | Liabilities-to-Assets (%) | Operating Cash / Net Income (x) |
|---|---|---|---|---|---|---|---|---|
| **2** | Apple | 416,161 | 112,010 | 111,482 | 26.92% | 26.79% | 79.48% | 1.00x |
| **5** | Microsoft | 281,724 | 101,832 | 136,162 | 36.15% | 48.33% | 44.51% | 1.34x |
| **8** | Tesla | 94,827 | 3,855 | 14,747 | 4.07% | 15.55% | 39.87% | 3.83x |

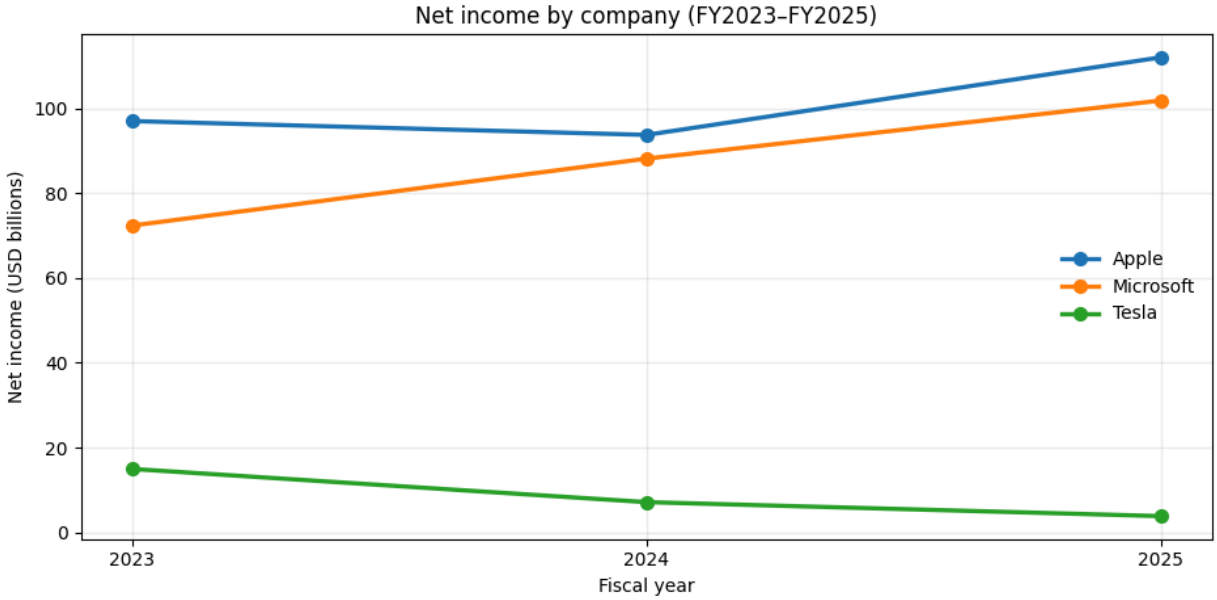
## 7 | Exploratory visualizations

The next four figures show **reported values**, not forecasts. Line charts are scaled to USD billions only for readability ( `USD m ÷ 1,000` ); the underlying data remains stored as USD millions.

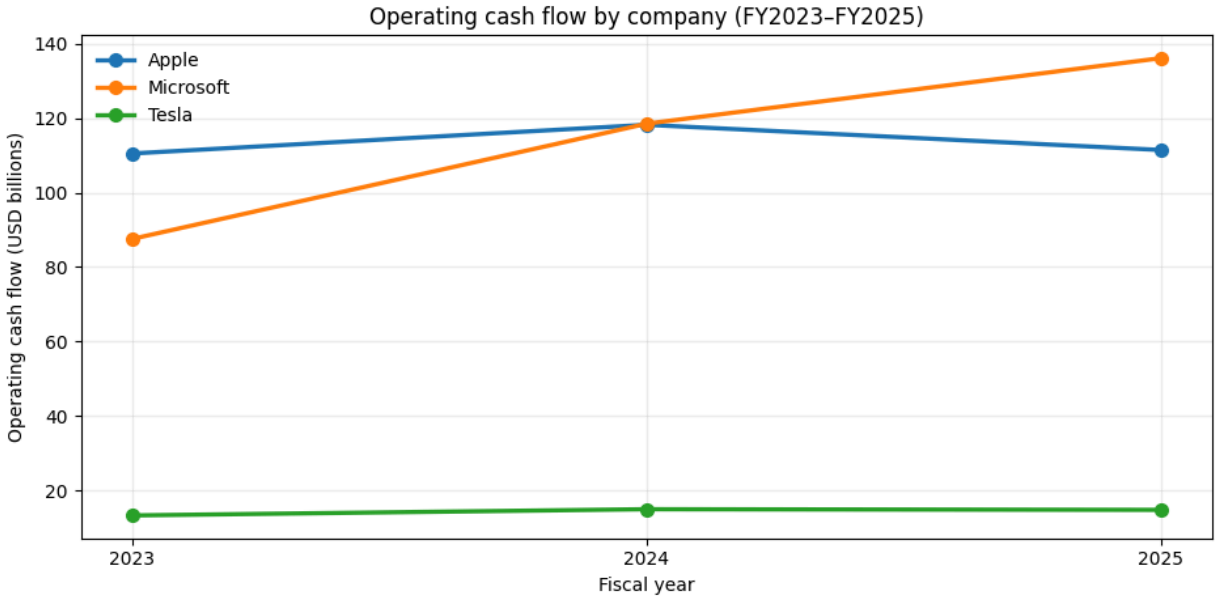
In [6]:
```
# Figure 1: annual revenue trend across the three companies
fig, ax = plt.subplots(figsize=(9.5, 4.8))
for company, g in df.groupby('Company'):
    ax.plot(g['Fiscal Year'], g['Total Revenue (USD millions)']/1000,
            marker='o', linewidth=2.4, markersize=7, label=company)
ax.set(title='Annual revenue by company (FY2023–FY2025)',
       xlabel='Fiscal year', ylabel='Revenue (USD billions)')
ax.set_xticks([2023, 2024, 2025]); ax.grid(alpha=.25); ax.legend(frameon=False)
fig.tight_layout(); plt.show()
```



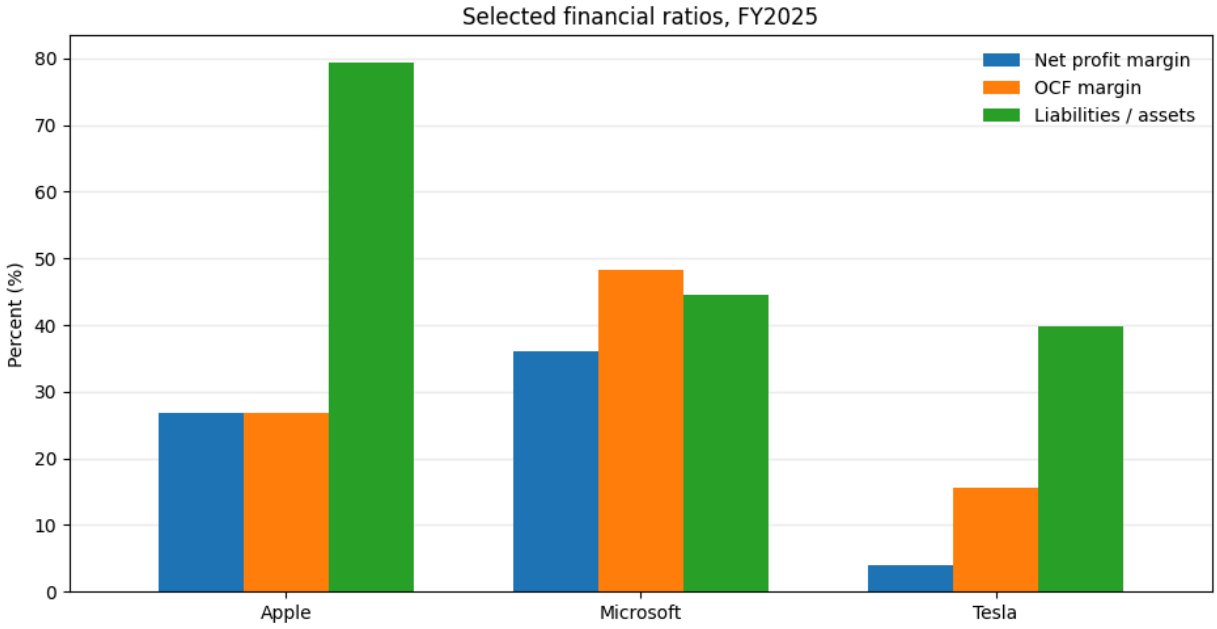

In [7]:
```
# Figure 2: net income trend
fig, ax = plt.subplots(figsize=(9.5, 4.8))
for company, g in df.groupby('Company'):
    ax.plot(g['Fiscal Year'], g['Net Income (USD millions)']/1000,
            marker='o', linewidth=2.4, markersize=7, label=company)
ax.set(title='Net income by company (FY2023–FY2025)',
       xlabel='Fiscal year', ylabel='Net income (USD billions)')
ax.set_xticks([2023, 2024, 2025]); ax.grid(alpha=.25); ax.legend(frameon=False)
fig.tight_layout(); plt.show()
```



In [8]:
```
# Figure 3: cash provided by operating activities
fig, ax = plt.subplots(figsize=(9.5, 4.8))
for company, g in df.groupby('Company'):
    ax.plot(g['Fiscal Year'], g['Operating Cash Flow (USD millions)']/1000,
            marker='o', linewidth=2.4, markersize=7, label=company)
ax.set(title='Operating cash flow by company (FY2023–FY2025)',
       xlabel='Fiscal year', ylabel='Operating cash flow (USD billions)')
ax.set_xticks([2023, 2024, 2025]); ax.grid(alpha=.25); ax.legend(frameon=False)
fig.tight_layout(); plt.show()
```




In [9]:
```
# Figure 4: comparative 2025 profit, cash-flow and balance-sheet ratios
indicators = ['Net Profit Margin (%)','Operating Cash Flow Margin (%)','Liabilities-to-Assets (%)']
compare = df[df['Fiscal Year'] == 2025].set_index('Company')[indicators].sort_index()
fig, ax = plt.subplots(figsize=(9.5, 5))
compare.plot.bar(ax=ax, rot=0, width=.72)
ax.set(title='Selected financial ratios, FY2025', xlabel='', ylabel='Percent (%)')
ax.grid(axis='y', alpha=.22); ax.set_axisbelow(True)
ax.legend(['Net profit margin', 'OCF margin', 'Liabilities / assets'], frameon=False)
fig.tight_layout(); plt.show()
```



## 8 | Evidence-based findings

### Microsoft | Strong and sustained growth

- **Revenue:** $211.9bn (FY2023) → $245.1bn (FY2024) → **$281.7bn (FY2025)**. Revenue expanded **14.93%** in FY2025 (and **32.94%** over the FY2023–FY2025 endpoints).
- **Net income:** $72.4bn → $88.1bn → **$101.8bn**; FY2025 growth **15.54%**.
- **Operating cash flow:** $87.6bn → $118.5bn → **$136.2bn**; FY2025 growth **14.86%**.
- **Balance sheet:** Assets increased to **$619.0bn** in FY2025; liabilities were **$275.5bn**. Liabilities/assets measured **44.51%**, a descriptive ratio rather than a borrowing-cost or solvency verdict.

**Interpretation:** Microsoft's 2025 results combine rising sales, profit, and operating cash generation. Continued financial monitoring should track whether cash generation keeps pace with growth.

### Apple | Sales and profits rose in FY2025, but operating cash flow fell

- **Revenue:** $383.3bn → $391.0bn → **$416.2bn**, up **6.43%** in FY2025.
- **Net income:** $97.0bn → $93.7bn → **$112.0bn**, up **19.50%** in FY2025 after declining in FY2024.
- **Operating cash flow:** $110.5bn → $118.3bn → **$111.5bn**, down **5.73%** in FY2025 despite the earnings rebound.
- **Balance sheet:** Assets decreased **1.57%**, and liabilities decreased **7.31%** in FY2025; the FY2025 liabilities/assets ratio was **79.48%**.

**Interpretation:** Sales and reported earnings improved, while annual operating cash inflow decreased from its FY2024 level. A financial chatbot should explain that cash flow and net income need not move together, without automatically classifying this as distress.

### Tesla | Profitability weakened despite asset expansion

- **Revenue:** $96.8bn → $97.7bn → **$94.8bn**, down **2.93%** in FY2025.
- **Consolidated net income:** $15.0bn → $7.2bn → **$3.9bn**; down **52.23%** in FY2024 and a further **46.11%** in FY2025.
- **Operating cash flow:** $13.3bn → $14.9bn → **$14.7bn**; a modest **1.18%** FY2025 decline, still positive.
- **Balance sheet:** Assets increased **12.89%**, while liabilities rose **13.54%** in FY2025. FY2025 liabilities/assets was **39.87%**.

**Interpretation:** Tesla expanded its reported asset base while sales and net income weakened. Positive operating cash flow partially offsets, but does not eliminate, concerns about declining profitability. The very high FY2023 income baseline should not be interpreted as a normalized earnings benchmark without reviewing tax and other one-off items in the 10-K notes.

### Overall comparison

Microsoft had the **fastest FY2025 revenue growth** of the three (+14.93%). Apple recorded the **largest absolute FY2025 revenue** ($416.2bn). Microsoft had the **highest FY2025 net profit margin** (36.15%), while Tesla recorded the **largest FY2025 net income decline**

(-46.11%). These are **historical observations**, not recommendations to buy or sell securities.

## 9 | Recommendations for the proposed GenAI financial chatbot

| Capability | Evidence from this exercise | Practical requirement |
|---|---|---|
| Reliable extraction | SEC 10-Ks contain inconsistent labels for the same concept ("net sales", "total revenues") | Map synonyms to a canonical metric and preserve the original statement label and SEC URL |
| Correct time comparisons | Apple, Microsoft and Tesla have different fiscal-year ends | Store `company`, `fiscal_year`, `fiscal_year_end`, `period_type` and `filing_date` with each observation |
| Provenance / trust | FY2023 balance sheets were taken from FY2024 filings | Return source citations and the **source filing year** with every numeric answer |
| Historical trend analysis | Some revenue, income and cash-flow trends diverge | Offer YoY percent changes, tables, trend charts and short explanations, with the reference years explicit |
| Safe, understandable interpretation | Apple OCF declined despite higher net income; Tesla cash flow stayed positive as earnings fell | Separate observed facts from inferred causes and avoid unsupported company-health labels |
| Scaling and quality | Manual extraction is slow and can contain copy errors | Introduce automated SEC/XBRL ingestion with human validation, missing-data flags and accounting-concept checks |

**Example end-user questions the chatbot should answer:**

- “How did Tesla's consolidated net income change between FY2024 and FY2025?”
- “Show Microsoft's revenue and operating cash flow from FY2023 to FY2025.”
- “Compare Apple's FY2025 net profit margin with Microsoft's, and cite the 10-Ks.”
- “Why should FY2025 results from these companies not be treated as identical calendar-year periods?”

**Expected answer structure:** metric, company, fiscal year and fiscal-year-end date, amount/unit, historical comparison, concise interpretation, and clickable SEC citation.

## 10 | Limitations and next steps

1. **Three annual points only:** These observations establish historical direction but are insufficient for dependable forecasting or inference about seasonality.
2. **Fiscal-year misalignment:** FY2025 periods end in June (Microsoft), September (Apple), and December (Tesla). They are not directly aligned calendar years.
3. **GAAP reporting and one-off effects:** Accounting policies, acquisitions, tax changes and nonrecurring items affect comparability. Detailed causal attribution requires further reading of management discussion and notes.
4. **Book balance-sheet measures:** Liabilities/assets reflects accounting book values. It is **not** the same as debt-to-equity, liquidity, credit rating or enterprise risk.
5. **Data scope:** Only five top-level metrics were manually extracted; full investment analysis also requires industry context, segment results, working capital, capex, debt maturities, and market risk.
6. **Live chatbot not yet built:** This notebook produces **clean, traceable analytics for a future model**. Developing a conversational application belongs to subsequent phases of the simulation.

**Next phase:** Ingest new 10-K/10-Q filings automatically, reconcile standardized XBRL tags, flag restatements and definition changes, expose transparent metric calculations via a structured retrieval layer, and evaluate chatbot answers against SEC ground truth.

## 11 | Export analysis results (optional)

The original manually extracted data is embedded in this notebook. If you run the following cell locally, it produces two optional companion CSVs in the current notebook folder: raw standardized observations and an enriched analysis table. **The notebook itself does not depend on those files** and can be uploaded alone to Forage.

In [10]:
```
# Reproducible exports: useful when passing the dataset to future AI/chatbot steps.
raw_columns = ['Company','Fiscal Year','Fiscal Year End'] + numeric_metrics + [
    'Financial Statement Source (10-K)', 'Balance Sheet Source (10-K)'
]
df[raw_columns].to_csv('BCG_GFC_Financial_Data_2023_2025.csv', index=False, date_format='%Y-
%m-%d')
df.to_csv('BCG_GFC_Enriched_Analysis_2023_2025.csv', index=False, date_format='%Y-%m-%d')
print('Created: BCG_GFC_Financial_Data_2023_2025.csv')
print('Created: BCG_GFC_Enriched_Analysis_2023_2025.csv')
print('Completed: 9 annual records, 5 raw metrics, 5 YoY measures and 4 financial
indicators.')
```

```
Created: BCG_GFC_Financial_Data_2023_2025.csv
Created: BCG_GFC_Enriched_Analysis_2023_2025.csv
Completed: 9 annual records, 5 raw metrics, 5 YoY measures and 4 financial indicators.
```